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ECEN 470
10/21/2026

PSO6:
5.1(a,b,c-i, c-ii, c-iii),
5.13(a-use MATLAB)(b)(c)(d)

1/2

(5.1) Set up the listed characteristic equations in the form suited to Evans root-locus method. Give $L(s)$, $a(s)$, and $b(s)$ and the parameter K in terms of the original parameters in each case. Be sure to select K so $a(s)$ and $b(s)$ are monic in each case, and the degree of $b(s)$ is not greater than that of $a(s)$.

a.) $s + (1/\tau) = 0$ versus parameter τ

b.) $s^2 + cs + c + 1 = 0$ versus parameter c

c.) $(s+c)^3 + A(Ts+1) = 0$

i.) versus parameter A

ii.) " " T

iii.) " " c, if possible. Say why you can or cannot. Can a plot of roots be drawn versus c for given constant values of A and T by any means at all?

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a.) $1 + K\frac{b(s)}{a(s)} = 0 \qquad 1 + \frac{1/\tau}{s} = 0, \quad 1 + \frac{1}{\tau}\frac{1}{s} = 0$

$K = \frac{1}{\tau}; \; b(s) = 1; \; a(s) = s; \; L(s) = \frac{b(s)}{a(s)} = \frac{1}{s}$

b.) $s^2 + cs + c + 1 = 0$

$s^2 + c(s+1) + 1 = 0$

$c(s+1) = -s^2 - 1$

$\frac{c(s+1)}{-s^2-1} = 1$

$\frac{c(s+1)}{-s^2-1} - 1 = 0$

$-\frac{c(s+1)}{s^2+1} - 1 = 0$

$1 + c\frac{(s+1)}{s^2+1} = 0$

$K = c; \; b(s) = s+1; \; a(s) = s^2+1; \; L(s) = \frac{s+1}{s^2+1}$

c.) $(s+c)^3 + A(Ts+1) = 0$

i.) versus A: $1 + AT\frac{(s+1/T)}{(s+c)^3} = 0$

$K = AT; \; b(s) = s + \frac{1}{T}; \; a(s) = (s+c)^3; \; L(s) = \frac{s+1/T}{(s+c)^3}$

ii.) versus T: $(s+c)^3 + ATs + A = 0$

$(s+c)^3 + A + (As)T = 0$

$1 + AT\frac{s}{(s+c)^3 + A} = 0$

$K = AT; \; b(s) = s; \; a(s) = (s+c)^3 + A; \; L(s) = \frac{s}{(s+c)^3 + A}$

iii.) versus c: NOT POSSIBLE: can't isolate c w/ c, c^2, c^3

Could solve numerically.

* See MATLAB

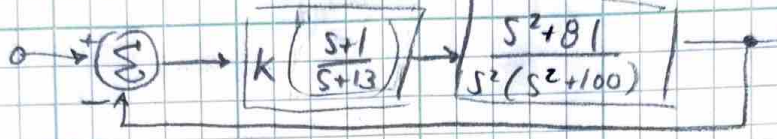
6.13 For system below, [2/2]

a.) Find locus of closed-loop roots w/ respect to K (use MATLAB)

b.) Is there val of K that causes all roots to have damping ratio greater than 0.5?

c.) Find values of K that yield closed-loop poles with damping ratio $\zeta = 0.707$

d.) Use MATLAB to plot response of the resulting design to a reference step



a.) Char eq: $1 + K\frac{(S+1)(S^2+81)}{S^2(S+13)(S^2+100)} = 0$ → see MATLAB for plot

b.) $\zeta = \sin\phi$

$\phi = \sin^{-1}\zeta = \sin^{-1}(0.5) = 30°$

$\phi \geq 30°$

There is NOT a value where $\zeta > 0.5$. This is because there are loci that don't cross that on the graph.

c.) $b(s) = (s+1)(s^2+81)$

$a(s) = s^2(s+13)(s^2+100)$

$1 + K\frac{b(s)}{a(s)} = 0$

dominant poles determined by magnitude and rate of decay.
↳ use graph

↳ $K \approx 88.2$ (w/ graph)

d.) See MATLAB

5.1 a.) iii.)

(s+c)^3 + A(Ts+1) = 0

Can a plot of the roots be drawn versus c for given constant values of and by any means at all?

```
%% Written by AI, edited by Seth Ricks

A = 1;
T = 1;

% Define a range of values for c to sweep through
c_values = linspace(0, 5, 100);

figure;
hold on;

for i = 1:length(c_values)
    c = c_values(i);

    poly_coeffs = [1, 3*c, 3*c^2 + A*T, c^3 + A];

    r = roots(poly_coeffs);

    % Plot the roots on the complex s-plane
    plot(real(r), imag(r), 'b.');
end

grid on;
xlabel('Real Axis');
ylabel('Imaginary Axis');
title('Root Locus-style Plot of Roots versus Parameter c');
hold off;
```

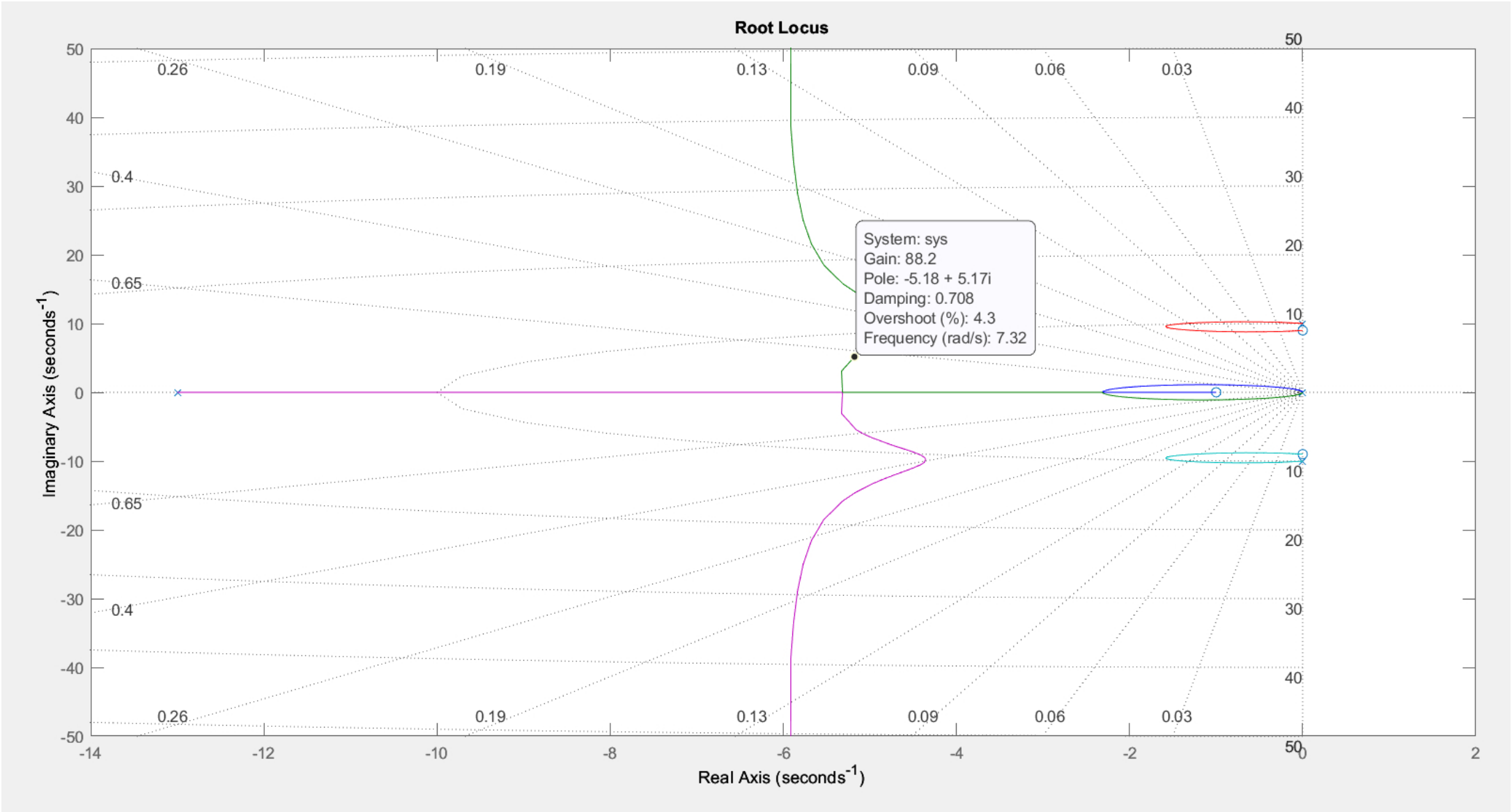

Root Locus-style Plot of Roots versus Parameter c
Imaginary Axis
Real Axis
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-6
-5
-4
-3
-2
-1
0
1

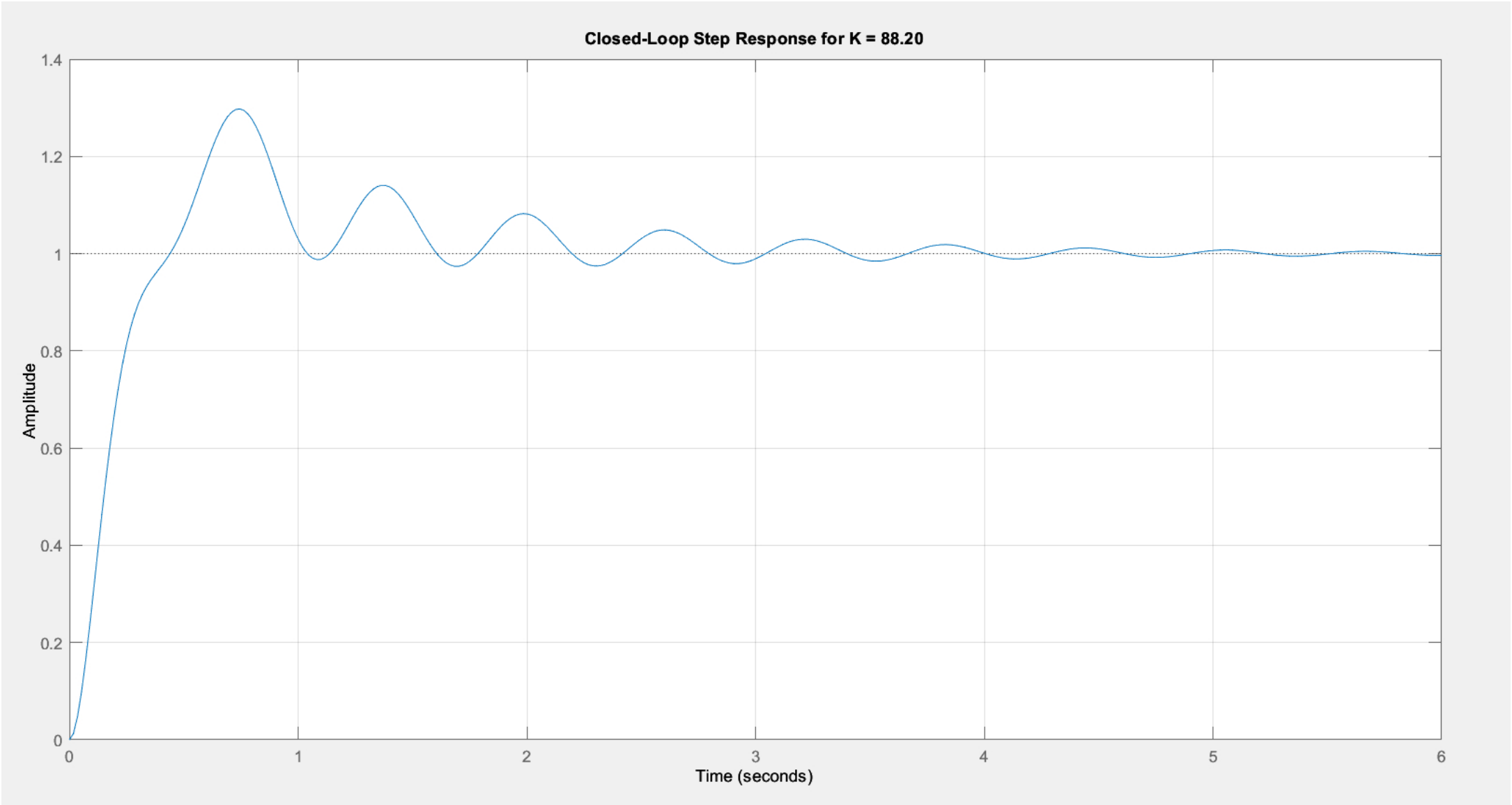
```
% Problem 5.13

clf;
s = tf('s');
sys = ((s+1)*(s^2+81)) / ((s+13)*(s^2)*(s^2+100));
K = 88.2;

fig_rl = figure;
rlocus(sys);
grid on;

fig_step = figure;
step(T);
grid on;
title(sprintf('Closed-Loop Step Response for K = %.2f', K));
```


Root Locus
System: sys
Gain: 88.2
Pole: -5.18 + 5.17i
Damping: 0.708
Overshoot (%): 4.3
Frequency (rad/s): 7.32
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
0.26
0.19
0.13
0.09
0.06
0.03
0.4
0.65
50
40
30
20
10
0
-10
-20
-30
-40
-50
-14
-12
-10
-8
-6
-4
-2
2


Closed-Loop Step Response for K = 88.20
Amplitude
Time (seconds)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
0
1
2
3
4
5
6